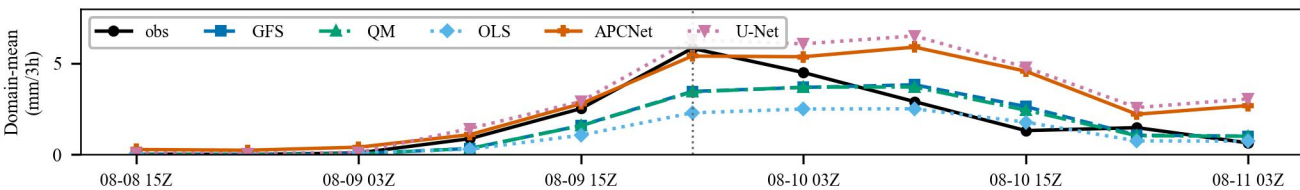


Typical Northeast China cold-vortex heavy-rain event: 2024-08-09 21Z (peak window, domain 25 x 37)

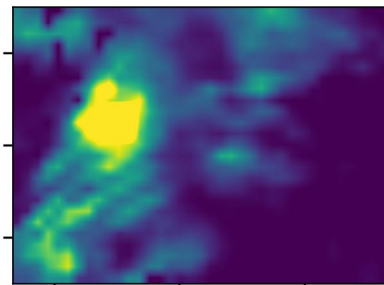
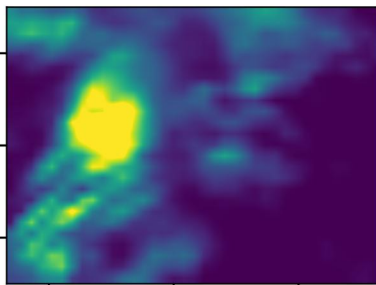
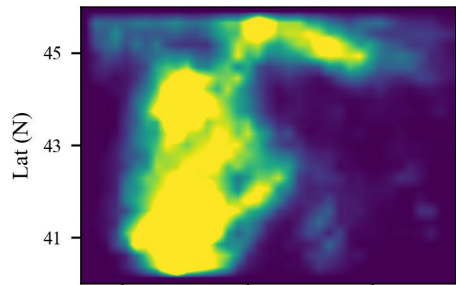
3-h domain-mean precipitation around the peak window



GPM IMERG (obs) (5.83)

GFS (raw) (3.47)

QM (3.44)



(a)

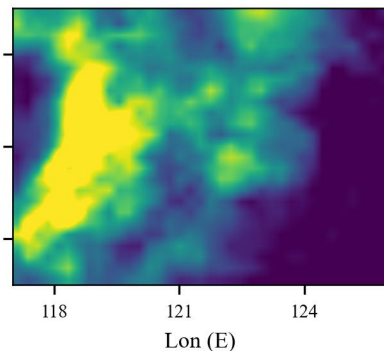
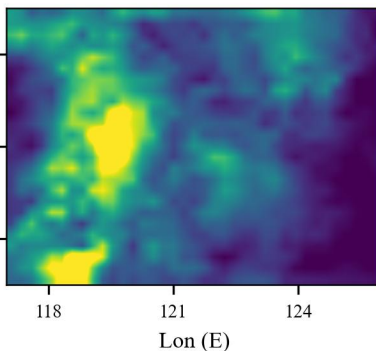
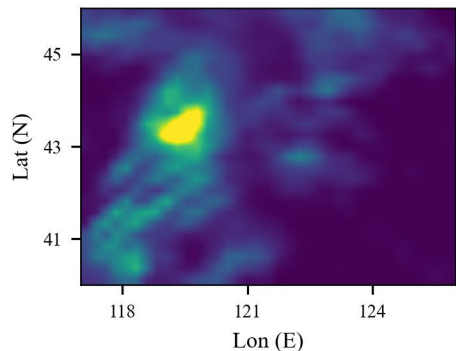
(b)

(c)

OLS (2.29)

APCNet (5.40)

U-Net (6.30)



(d)

(e)

(f)

